

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)			<u>Radiation</u> may come from different directions or to monitor <u>radiation</u> levels across Wales or background <u>radiation</u> may vary across Wales		1		1		
	(b)			Measure count for a lengthy period of time e.g. 5 minutes (1) Divide the count value by the time to get cpm (1) Alternative: Measure count for 1 minute (1) Repeat this numerous times and calculate a mean to get cpm (1)	2			2		2
	(c)	(i)		It is the time taken to halve (1) number of radioactive particles or nuclei or atoms / activity / mass / amount of substance / count rate (1)	2			2		
		(ii)		8 [days]		1		1	1	
	(d)			$[1 \rightarrow] \frac{1}{2} \rightarrow \frac{1}{4} \rightarrow \frac{1}{8} \rightarrow \frac{1}{16} \rightarrow \frac{1}{32} \rightarrow \frac{1}{64} \rightarrow \frac{1}{128} \rightarrow \frac{1}{256} \rightarrow \frac{1}{512}$ (1) 9 [half-lives] (1) The first mark can be implied if 9 half-lives shown so two marks can be awarded. $9 \times 30.2 = 271.8$ (1) [years] Alternative: $2^n = 512$ (1) $n = 9$ (1) $9 \times 30.2 = 271.8$ (1) [years]		3		3	2	

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	(e)			I-131 has a shorter half-life [than Cs-137] (1) accept converse and nearly all of it will have decayed [in 35 years] (1) Alternative: Cs-137 will not be safe for 271.8 years or the activity of Cs-137 will still nearly be a half [after 35 years] (1) I-131 will be safe after 72 days (1)		2		2		
				Question 6 total	4	7	0	11	3	2